l-Diversity for Top-Down Greedy

Extending the existing k-anonymity implementation with distinct l-diversity

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SPAI 2026, Exercise 1.2 (Anonymisation)

Why l-diversity?

- k-anonymity requires at least k records to share the same quasi-identifiers.
- But all records in an equivalence class may still share the same sensitive value.
- This can reveal sensitive information even though the data is k-anonymous.

Our goal:
enforce both $|EC| \geq k$ and
at least l distinct sensitive values

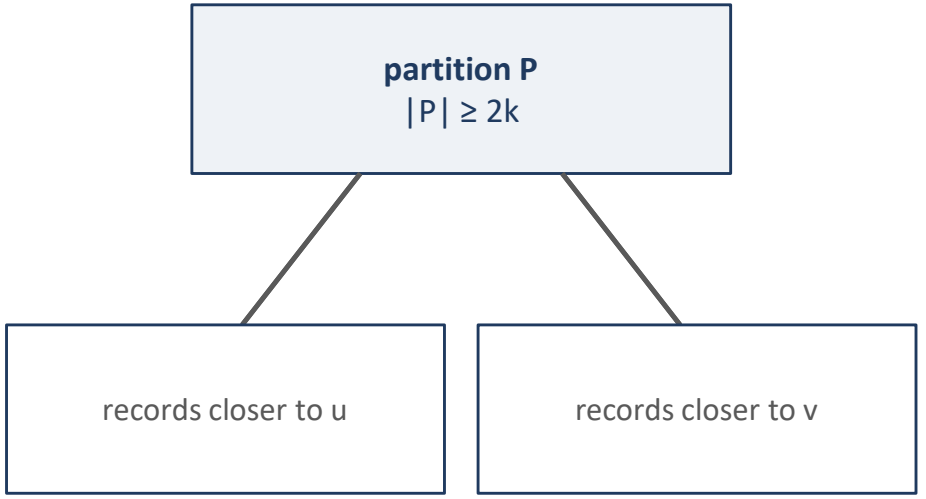
Sensitive attribute:
ocean_proximity

Dataset:
cahousing

Evaluate the privacy–utility trade-off using information loss and classification performance.

Top-Down Greedy (baseline)

- Start with the full dataset as one partition.
- Choose two distant seed records using NCP-based distance.
- Assign each record to its closer seed.
- Balance partitions if one side contains fewer than k records.
- Recurse until no further valid split is possible.
- Generalize each final partition.



Only QIs drive the split.

The sensitive attribute is ignored.

Our change: validate and retry candidate splits

1

Split

Generate a normal TDG candidate split.

→

2

Balance

Apply the original k-anonymity balancing step.

→

3

Validate

Both children must satisfy k and distinct l-diversity.

→

4

Retry / stop

Try up to 10 candidates; otherwise keep the parent partition.

Initial attempt

reject the first invalid split → one giant EC (NCP = 1.000)

Final implementation

retry up to 10 candidate splits → much lower information loss

- Sensitive attribute index is recalculated after the framework reorders columns.
- random.seed(42) makes the anonymization results reproducible.

Experimental setup

- cahousing: 20,640 records
- 5 QIs: longitude, latitude, housing_median_age, median_income, median_house_value
- Sensitive attribute / classification target: ocean_proximity (3 values)
- Configurations: k = 2, 5, 10; l = 2, 3 where applicable
- Same predefined train/validation split for every configuration.
- Generalized numeric ranges are represented by their midpoint for classification.

Information loss

NCP • CAVG • DM

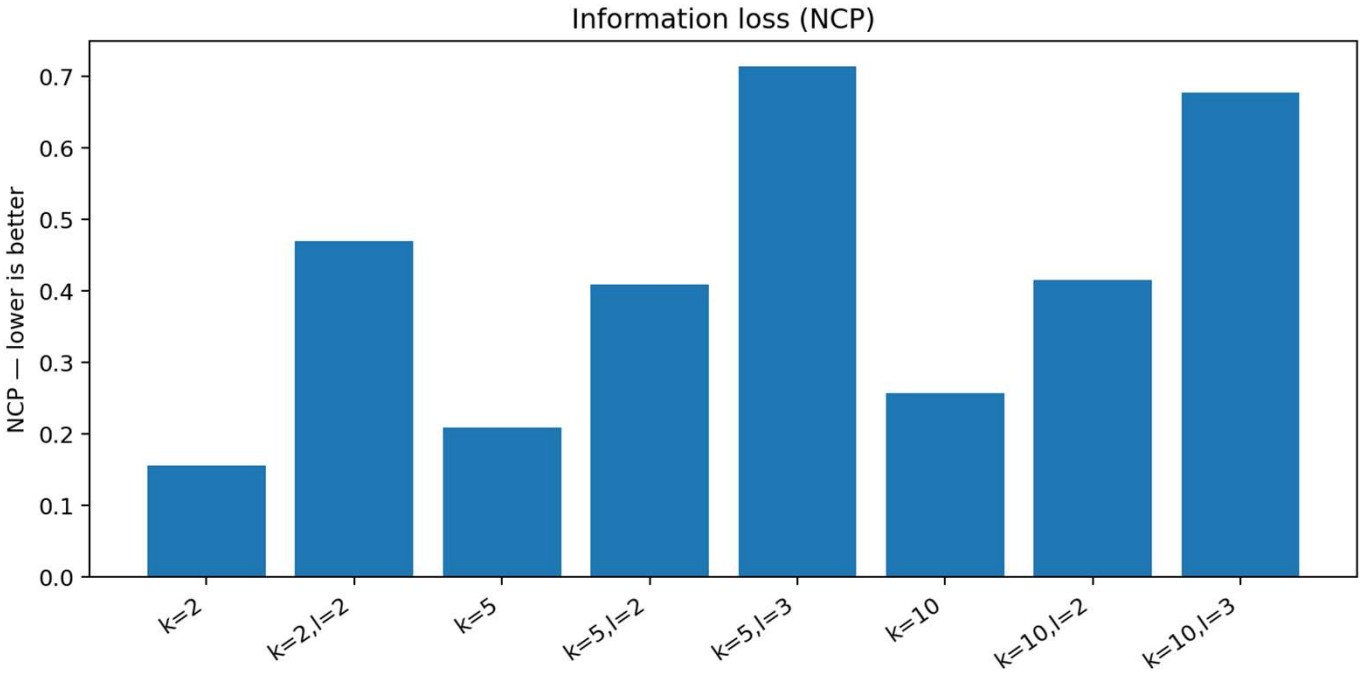
Classification utility

Random Forest • weighted F1

Privacy validation

EC size • min. l - violation counts

Privacy is satisfied — but stronger l increases information loss



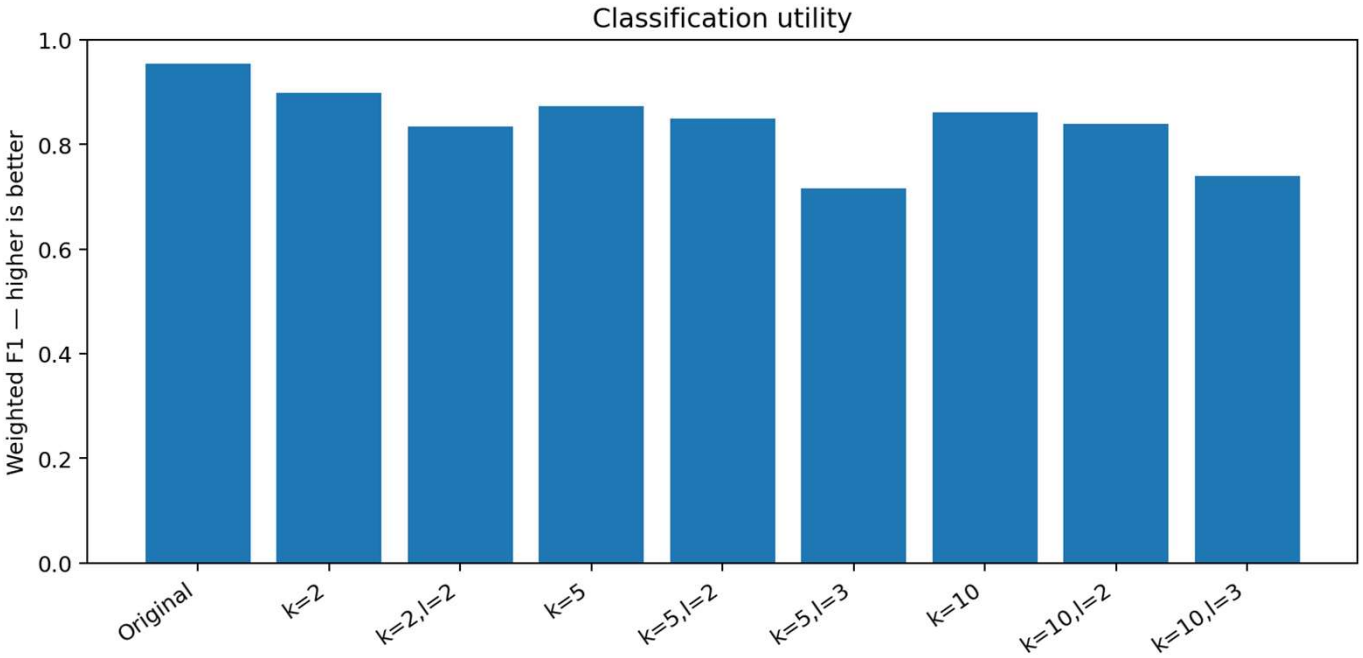
0 k-anonymity violations in every run

0 l-diversity violations in every l-diverse run

k=5:
NCP 0.208 → 0.408 → 0.713
as l increases

l has a stronger utility cost than increasing k alone.

Classification utility decreases as privacy strengthens



Original data

F1 = 0.954

TDG k=5

F1 = 0.874

TDG k=5,l=3

F1 = 0.716

Distinct l=3 has the largest classification penalty.

Conclusion

- The TDG extension successfully enforces distinct l-diversity on top of k-anonymity.
- Retrying alternative candidate splits is much more useful than rejecting the first invalid split.
- Increasing k causes moderate information loss; increasing l has a substantially stronger effect.
- The strongest privacy settings also produce the largest drop in Random Forest performance.
- The k/l effect is not perfectly monotonic because TDG is heuristic and depends on the resulting partition structure.

Main takeaway

Stronger diversity guarantees improve privacy, but at a measurable cost in data utility.

Thank you! Questions?